B.VOC (HARDWARE AND NETWORKING) 2016-2017

Scheme

Semester-I

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
***GEN-101**	Communication Skills	Generic	Theory	20	80	-	-	6
***GEN-102**	Fundamentals of Information Technology	Generic	Theory	20	80	-	-	6
HWN-103	Computer Hardware	Skill	Theory & Practical	10	40	10	40	6
HWN-104	Network Fundamentals	Skill	Theory & Practical	10	40	10	40	6
HWN-105	Practical on Networking-I	Skill	Practical	-	-	20	80	6

Semester-II

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
***GEN-201**	Soft Skills and Personality Development	Generic	Theory	20	80	-	-	6
GC-202	Business Ethics	Generic	Theory	20	80	-	-	6
HWN-203	Personal Computing Software	Skill	Theory & Practical	10	40	10	40	6
HWN-204	Computer Hardware and Trouble Shooting	Skill	Theory & Practical	10	40	10	40	6
HWN-205	Practical on Networking-II	Skill	Practical	-	-	20	80	6

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester-III

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-301	Value Education And Human Rights	Generic	Theory	20	80	-	-	6
GC-302	E-Commerce	Generic	Theory	20	80	-	-	6
HWN-303	Windows Operating System	Skill	Theory & Practical	10	40	10	40	6
HWN-304	Routing and Switching	Skill	Theory & Practical	10	40	10	40	6
HWN-305	Routing Configuration & Network Connectivity	Skill	Practical	-	-	20	80	6

Semester-IV

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-401	Environmental Studies	Generic	Theory	20	80	-	-	6
GC-402	Project Management	Generic	Theory	20	80	-	-	6
HWN-403	Windows Server Administration	Skill	Theory & Practical	10	40	10	40	6
HWN-404	Network Administration & Trouble Shooting	Skill	Theory & Practical	10	40	10	40	6
HWN-405	Switching Configuration & Network Connectivity	Skill	Practical	-	-	20	80	6

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester-V

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-501	Critical Thinking And Elementary Statistics	Generic	Theory	20	80	-	-	6
GC-502	Introduction To Research Methodology AndReport Writing	Generic	Theory	20	80	-	-	6
HWN-503	Linux Operating System	Skill	Theory & Practical	10	40	10	40	6
HWN-504	Firewall Technology	Skill	Theory & Practical	10	40	10	40	6
HWN-505	Advanced Routing and Switching	Skill	Practical	-	-	20	80	6

Semester-VI

Paper Code	Title	Generic/ Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credit
*GEN-601	Entrepreneurship Development Programme	Generic	Theory	20	80	-	-	6
GC 602	Total Quality Management	Generic	Theory	20	80	-	-	6
HWN-603	Industrial Training	Skill	Practical	-	-	-	300	18

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

SEMESTER-I

Paper Title: HWN-103-COMPUTER HARDWARE

Job Role: Network Support Technician

Theory:40
Internal Assessment: 10
:Total Marks: 50
Time:3 hours

Objectives: Hardware Fundamentals enable the students to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Introduction to computer hardware: Peripheral devices of a Computer system, Add On cards: network interface card, sound card and graphics card, functional description of various parts of a PC, UPS, Types of UPS: offline, online and line interactive UPS

UNIT-II

Various Components of a PC: Mother Board: Types of Motherboard AT, ATX, and BTX. Ports: Types of Ports, serial and parallel ports. Hard Disk: Types of Hard Disk: PATA, SATA, SCSI. RAM: Types of RAM: SRAM, DRAM, SDRAM,Power supply unit, cabinet,Processor.

UNIT-III

Cables: Types of Cables: USB, VGA, DVI, RJ11, RJ45, HDMI, SERIAL, and PARALLEL. Connecting Cables from SMPS to motherboard, hard disk etc. Establishing data connection for mother board, hard disk, and drivers. Fixing wires for power restart switches, fixing wires for power & HDD LED, External USB and Audio Connections. Drivers: Types of drivers: Device drivers, LAN drivers, sound drivers, graphics drivers.

UNIT-IV

BIOS: Introduction, Connecting & disconnecting computer peripherals and componentsMouse, Keyboard, Monitor, Hard Disk. Window installation, Hard disk: partitioning and formatting, creating, formatting and deleting partitions using DISKPART in Command Prompt and Graphical User Interface. Antivirus: features and Significance, Installing third party application. Firewall: Introduction to firewall, Features of firewall.

Practical based on HWN-103

Practical:40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Govindarajalu, B., IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Tata McGraw Hill, 2nd ed. N.D.
- Basandra, S. K., Computers Today, Galgotia Publications
- Craig Zacker., The Complete Reference: PC Hardware, TMH Publication
- Scott Mueller., Upgrading and Repairing PC, Pearson
- Jean Andrew., A+ Guide to Hardware, 6th Edition

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Paper Title: HWN-104-NETWORK FUNDAMENTALS

Job Role: Network Support Technician

Theory:40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Objectives: Networking Fundamentals teaches the basic concepts and terminology of networking. The syllabus covers media types and standards. Students are also introduced to the terminology and basic concepts and Open Systems Interconnection (OSI) model.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising two questions from each unit, including Question No.1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Network concepts: Introduction to Network, characteristics, need, types of networks on the basis of area coverage: LAN, WAN, MAN; Advantages and Disadvantages of networks; Network Topologies: bus topology, ring topology, star topology, tree topology, mesh topology and hybrid topology. Network Models: Peer-to-Peer, protocol, client/server network and hybrid type network. Servers: Different types of servers, Network protocols

UNIT-II

Introduction to network standards: OSI reference model, TCP/IP Model. Difference between OSI reference model & TCP/IP model, Transmission Media: Types, Guided Media: twisted pair cable, coaxial cable, fibre optics, Unguided Media: radiowave, microwave, lightwave, infrared and satellite.

UNIT-III

Networking protocols: DNS, FTP, Telnet, DHCP, HTTP, TFTP, SSH, SFTP, SMTP, POP, IMAP, NTPNetwork devices: HUB, Switch, Repeater, Bridges, Routers, Gateway: IEEE Standards: IEEE Standards for LAN and WAN: IEEE Standard 802.3 and Ethernet, IEEE Standard 802.4 Token Bus, IEEE Standard 802.5 Token Ring, IEEE Standard 802.6 Distributed Queue Dual Bus, IEEE Standard 802.2 Logical Link Control.

UNIT-IV

IP address: Introduction, IP address format,classes of IP Address, subnet address, Configuration of IP Address in PC, Configuration of IP address in Switch, configuration of Wireless Devices, Access Points Network Interface card (NIC): working principle of NIC, MAC Address, User Management: overview of user account, managing user account and password security: adding user, setting and changing password, changing picture of user, deleting the user, setting account restriction for the user, renaming a local account.

Practical based on HWN-104

Practical:40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question

- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Tannenbaum, A.S., 2003: Computer Networks, Prentice Hall.
- Networking: The Complete Reference 1st Edition, Mcgraw Hill Education
- Stallings, William, Local and Metropolitan Area Networks: Macmillan Publishing Co.
- Black: Data Network, Prentice Hall of India.
- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

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Paper Title: HWN-105-PRACTICAL ON NETWORKING-I

Job Role: Network Support Technician

Practical: 80
Internal Assessment: 20
Total Marks: 100
Time:3 hours

Objectives: The objective of this paper is to provide in-depth knowledge about the concepts of networking. It offers integrated and comprehensive coverage of networking tools and devices.

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 10 marks.
- Practical file carries 10 marks and 40 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT-I

Networking Tools: Crimping tool, punching tool: Impact and Punch down Tool, Multi Punch Tool, RJ45 Punch Down Tool, Impact Tool, Computer Punch Tool – deluxe, Cable cutter, I/O device, I/O Face Plate, I/O Box, LAN Cards, Connectors: RJ-11 (Registered Jack), RJ-45 (Registered Jack), F-Type, ST (Straight Tip) and SC (Subscriber Connector or Standard

Connector), Fiber LC (Local Connector), MT-RJ (Mechanical Transfer Registered Jack), USB (Universal Serial Bus), LAN Patch Panel, Network Switch Rack

UNIT-II

Testing and Networking Tools: Introduction, Types of tools: LAN Testers, Application tools: IP scanner, Test Network Speed, Bandwidth Monitor 2 Lite Edition, Netbrute Scanner, piecework's IT Desktop, Ping, ipconfig, fstat, iproute, route, netstat, ipconfig /release, ipconfig /renew

UNIT-III

Working of cables: Coaxial cable, Unshielded twisted pair (UTP), Types of UTP Cable, Shielded Twisted-Pair Cable, Fiber optics, Patch Cords: UTP Patch cords, Fiber patch cords, Cable Manager, Cable Tag, Cable stripper, Scissors.

UNIT-IV

Working on Hub and Switch: Types of Hub: Active Hub, Passive Hub and Intelligent Hub, Types of switches: Unmanaged Network Switches, Managed Switches, Smart Switches.Setting up of network: UTP cables, Switch, LAN tester, Switch Rack, Patch Panel, Cable Manager, Cable tagging, I/O box, I/O face plate, patch cords, connectors, terminals.

Books Recommended:

- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Cisco Networking Essentials, Author: mcmillan, troy|author; John Wiley & Sons
- Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
- Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
- Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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SEMESTER-II

Paper Title: GC202 - BUSINESS ETHICS

Theory 80
Internal Assessment: 20
Total Marks: 100
Time:3 hours

Objective: The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Unit 1

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organisations, Theories of Business Ethics.

Unit-II

Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics. Ethical Issues in Indian Business.

Unit - III

Professional Values:Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality.

Unit - IV

Corporate Social Responsibility**:** Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit,Corporate Social Responsibility and Corporate Governance.

Suggested Readings:

1. J.P. Sharma, *Corporate Governance, Business Ethics & CSR,* Ane Books Pvt. Ltd., Nee Delhi.
2. Andrew Crane, Dirk Matten, *Business Ethics,* Oxford University Press, New Delhi.
3. Daniel Albuquerque, *Business Ethics, Principles and Practices (Indian Edition),* Oxford University Press, New Delhi
4. Fr. Floriano C. Roa, *Business Ethics and Social Responsibility*, Rexestore.
5. O. C. Ferrell, John Fraedrich, Linda Ferrell, *Business Ethics: Ethical Decision Making & Cases, Cengagae Learning*
6. Michael Blowfield, Alan Murray, *Corporate Responsibility – A Critical Introduction,* Oxford University Press, New Delhi.

Paper Title: HWN-203-PERSONAL COMPUTING SOFTWARE

Job Role: Network Support Technician

Theory: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

Objectives: The objective of this course is to familiarize students with Fundamentals of Information Technology and its applications. It enables the student to get practical exposure towards MS- Office tools

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Unit- I

DOS : Versions of DOS: Booting sequence; Warm and Cold reboot; Concept of File and directory , Redirecting command input and output pipes, Wildcard characters, Types of DOS commands: Internal and External; Internal Commands: DIR, MD, CD , CLS, COPY, DATE, DEL, PATH, PROMPT, REN, RD, TIME, TYPE, VER, VOL; External Commands: XCOPY, ATTRIB, BACKUP, RESTORE, FIND, SYS , FORMAT, CHKDSK, DISKCOPY, LABEL, MOVE, TREE, DELTREE, DEFRAG, SCA NDISK, UNDELETE.

Batch Files: Introduction to simple batch files; Introduction to CONFIG.SYS and AUTOEXEC.BAT files. Graphical User Interface: Fundamentals of windows, types of windows, anatomy of windows, Icons, Recycle bin Operations on window: Opening a Window, Minimizing and Maximizing a window, Moving window, Resizing Window, Closing the window windows explorer Folders: Creating and deleting folders, copying, renaming folders, folder properties. Control panel.

Unit- II

Word Processing Package: Basics of Word Processing; Opening and Closing of documents; Text creation and Manipulation; Finding and replacing text, Printing of document, Formatting of text; Margin setting, Adding Borders and shading, Adding Headers and Footers, Setting up Multiple columns, Working with tables, Spell check, Grammar facility, Auto text, language setting and thesaurus; Mail merging. Installation of Word Processing Software.

Unit- III

Spreadsheet Package : Worksheet Basics, Data Entry in Cells : Entry of numbers, text and formulae, Moving data in a worksheet, Moving around in a worksheet, Selecting Data Range, Using the Interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing; Formatting and Calculations: using Auto fill, Working with Formulae, Efficient Data Display with Data formatting (number formatting, date formatting etc.), Working with Ranges, Worksheet Printing; Working with Graphs and Charts : Creating Embedded Chart using char wizard, sizing and moving parts, updating charts, Changing chart types, Chart wizard, Adding Titles, Legends and Gridlines, Printing Charts; Database Management. Finding records with Data form, Adding/Deleting Records, Filtering Records in a worksheet; Functions and Macros: Worksheet Creating Macros, Recording Macros, Running Macros, Assigning Macros to Buttons, Defining Macros from Scratch. Multiple Worksheets. Installation of Spreadsheet software.

Unit- IV

Presentation Packages: Basics, General Features, Creating a presentation, formatting and enhancing text, Incorporation of Animation, adding charts, multimedia, page setup and printing slides. Installation of Presentation software. Internet and WWW: Evolution of Internet, services provided on Internet, Access Methods, application of Internet.

Practical based on HWN-203

Practical: 40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.

- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

Mathur Rajiv, 1996: Learning Word 6 for Windows Step-by-Step, Galgotia.
Mathur Rajiv, 1996: Learning Excel 5 for Windows Step-by-Step, Galgotia.
Jamsa, Kris A., 1993: Rescued by Windows 3.1, Galgotia. 5. Basandra, S.K., 1995 Computers Today, Galgotia.
Kasser, Barbara, 1998: Using the Internet, PHI, 4th ed., New Delhi.
Wall, David A. & Others, 1996: Using the World Wide Web, PHI, 2nd ed., New Delhi.

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Paper Title: HWN-204-COMPUTER HARDWARE AND TROUBLE SHOOTING

Job Role: Network Support Technician

Theory:40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Objectives: This paper enables the students to identify the hardware in-depthand helps them to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Unit- I

Trouble Shooting Basics: Basic Trouble shooting, the need for trouble shooting plan, elements of trouble shooting, general trouble shooting process. Diagnostic, Trouble Shooting and Repair Tools: Diagnostic software, Diagnostic hardware, the PC toolkit.Keyboard: Types of keyboard, correcting problems, vacuum cleaners and keyboards, keyboard maintenance, Mouse: Mouse basics, working with a mouse, performing mouse actions, inside the mouse, adjusting mouse properties, mouse keys, mouse maintenance, Optical mouse, breakdown and break through.

Unit- II

Monitors: basic, VGA, DVI, viewable area, maximum resolution and dot pitch, power consumption, flat panel, monitor resolution, colour depth, degaussing, power option for monitor, display option for monitor, monitor maintenance. Hard Disk Drives: History, construction and operations of hard disk, read/write heads, File allocation table (FAT, FAT12, FAT16), Virtual FAT (VFAT), 32-Bit FAT (FAT32), New Technology File System (NTFS), High Performance File System (HPFS), Hard Disk tools: clean-up, error checking, disk fragmentation, change drive name and letters, virus scan, formatting a hard disk from windows, formatting a hard disk from DOS windows, installing master slave drives, CMOS identification of master slave disk, breakdown and break through.

Unit- III

Processor: Basics, performance factors, modes, instruction set, CISC, RISC, and Superscalar. Processor associates: clock, motherboard, system bus, cache memory, instruction pipelining. Printers: printer characteristics, type quality, print speed, impact vs nonimpact, and fonts. Inkjet printers: common features, operations, parts of inkjet printer, control panel, trays, cartridges, cleaning cartridges, replacing cartridges, clearing paper jams. Laser Printers: common features, operations, technology, control panel, trays, output bin, cartridges, redistributing toner, internal cleaning, paper jams, cancel printing jobs, setting preferences and defaults of printing, breakdown and break through.

Unit- IV

Mother board: Introduction, types, processor sockets and slots, memory sockets, cache and/or cache sockets, I/O bus slots, power connector, voltage regulator, capacitors, keyboard and mouse connectors, chipset chips, keyboard controller, real-time clock, super I/O controller, BIOS chips, battery, jumpers, ports and headers, pin connectors, upgrading motherboards, installing RAM and new processor. Power and SMPS: Power supply, AC-DC voltage conversion, parts of power supply, trouble shooting

Practical based on HWN-204

Practical: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Govindarajalu, B., IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Tata McGraw Hill, 2nd ed. N.D.
- Basandra, S. K., Computers Today, Galgotia Publications
- Craig Zacker., The Complete Reference: PC Hardware, TMH Publication
- Scott Mueller., Upgrading and Repairing PC, Pearson
- Jean Andrew., A+ Guide to Hardware, 6th Edition

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Paper Title: HWN-205-Practical on Networking–II

Job Role: Network Support Technician

Practical :80
Internal Assessment: 20
Total Marks: 100
Time:3 hours

Objectives: The objective of this paper is to provide in-depth knowledge about the concepts of networking. It offers integrated and comprehensive coverage of networking tools and devices.

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 10 marks.
- Practical file carries 10 marks and 40 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Unit- I

Patching of LAN Cable: Strip cable end, Untwist wire ends, Arrange wires, Trim wires to size, Attach connector, Check, Crimp, and Test. Setting up of LAN: UTP cables, Switch, LAN tester, Switch Rack, Patch Panel, Cable Manager, Cable tagging, I/O box, I/O face plate, patch cords, connectors, terminals.

Unit- II

Internet connectivity: Different types of internet connection: Wireless, mobile, Hotspot, Dial-up, Broadband, DSL, Cable, Satellite, ISDN, Internet Service Provider (ISP), bandwidth or speed tester, Internet connectivity switch and PC, configuration of switch, managing of switching, troubleshooting of switching, testing application for switch.

Unit- III

Wi-fi Connectivity: Basic Configuration of Wi-Fi device or access point, IP address configuration, DNS Address, DHCP mode, Security and Password, Admin login and password, MAC Filtering, Monitoring, Backup and restore, etc.

Unit- IV

Switch connectivity: Managed and unmanaged switch, Basic configuration of switch, mac filtering, IP address configuration, DNS address, DHCP Mode, Admin login and Password, VLAN, backup file and restore settings.

Books Recommended:

- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
- Tannenbaum, A.S., 2003: Computer Networks, Prentice Hall.
- Networking: The Complete Reference 1st Edition, Mcgraw Hill Education
- Stallings, William, Local and Metropolitan Area Networks: Macmillan Publishing Co.
- Black: Data Network, Prentice Hall of India.

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SEMESTER-III

Paper Title : GC 302 –E-COMMERCE

Theory:80
Internal Assessment: 20
Total Marks : 100
Time:3 hours

Objective : The objective of this paper is to provide fundamental knowledge to the students about E-Commerce so that they can better perform in any area of operation and can excel in the field of commerce with IT specialization.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Electronic Commerce Framework, Basics and Tools of E-Commerce, Comparison of Web-based with Traditional Business; Growth of E-Commerce – Present and potential. E-Business: Meaning, Importance, Models Based on the Relationships of Transacting Parties (B2B, B2C, C2C and C2B), Present Status of E-Commerce in India.

UNIT II

Changing Structure of Organization – The Impact of E-Commerce on supply chain management and customer relation, Socio-Economic Impacts of E-Commerce. Electronic Payment System: Types of Payment System — E-Cash and Currency Servers, E-Cheques, Credit Cards, Smart Cards, Electronic Purses and Debit Cards , Risk and Electronic payment Systems.

UNIT-III

Electronic Data Interchange (EDI): EDI and its applications in business; Legal, Security and Privacy issues in EDI, EDI standardization; Internet based EDI, Internal Information Systems, Work-flow Automation and Coordination, Types of Digital Documents, Digital Signatures.

UNIT-IV

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP, ERP Modules and design alternatives, challenges of ERP implementation, Business Process Re-engineering, ERP system development process. Regulatory aspects of E-commerce.

Books Recommended:

- 'Electronic Commerce, A Manager's Guide to E-Business' by Prag Diwan and Sushil Sharma, Vanity Books International, Delhi.
- 'Frontiers of Eletronic Commerce' by Ravi Kalakota, and Andrew B.Shinston, Addision Wesley.
- 'Web Commerce Technology Handbook' by Minoli and Minoli, Tata McGraw Hill, New Delhi.
- 'Electronic Commerce' by Gray P.Schneider, Course Technology, Delhi.
- 'E-Commerce-The Cutting Edge of Business' by Kamlesh Bajaj & Debjani Nag, Tata Mc Graw Hill 2000.
- 'Understanding E-Commerce' by David Kosiur, Microsoft Press. 1997

Paper Title: HWN-303-WINDOWS OPERATING SYSTEM

Job Role: Network Support Engineer

Theory:40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objective: This course enables students to understand the concepts of Operating System. After the completion of this paper, student will be able to manage various processes, files and folders on the disk and desktop with effective outcome.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.
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UNIT- I

Operating System: Introduction, features, Windows Operating System: features, history, hardware requirements, And installation of Windows Operating SystemGraphics interface: Benefits, Screen attributes: icons and bars. Mouse vs. keyboard input Features and accessories of the Windows program. Objects and their properties. Program management: Creating, saving and editing documents, Concurrent execution of programs, sharing data between programs.

UNIT- II

Folder and file management: Working with files, Naming files, Navigate to Folders with Windows Explorer, Copying and moving files, Deleting files, Managing folders, Creating,

Viewing, Expanding and collapsing, Backing up and restoring files. Components of Windows: format of a window, moving windows, resizing windows, minimizing and maximizing windows.

UNIT- III

Control panel: Customizing screens, Screen colors, Patterns, Spacing icons, selecting time/date, setting the Sound, Concept of menu Using Help, Creating Short cuts, Basics of Window Setup, Notepad, Window Accessories, System restore.Customizing printing, changing the print queue, configuring the printer, Adding printers. Working with fonts: changing, removing, adding, customizing mouse and keyboard use.

UNIT- IV

System properties and the device manager Management tools, DOS sessions, Explorer, Memory configuration, Safe mode Install and uninstall applications, Setup/troubleshooting issues.Maintaining and optimizing disks: Disk Cleanup, Disk defragmenter, Compressing and uncompressing folders and files. Encrypting and decrypting folders and files.

Practical based on HWN-303

Practical: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Jamsa, Kris A., Rescued by Windows 3.1, Galgotia.
- Basandra, S.K., Computers Today, Galgotia.
- Mathur Rajiv, Learning Word 6 for Windows Step-by-Step, Galgotia.
- Mathur Rajiv, Learning Excel 5 for Windows Step-by-Step, Galgotia.
- Darril Gibson, Microsoft Windows Networking Essentials 1st Edition.

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Paper Title: HWN-304-ROUTING & SWITCHING

Job Role: Network Support Engineer

Theory:40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objective: This course provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, it enable the students to enhance career development skill in the field of networking.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT- I

Router: Introduction, components of routers, Types of Routers: Broadband Routers, Wireless Routers, Edge Router, Subscriber Edge Router, Inter-provider Border Router, Core Router, Wired and Wireless Routers. Functions of routers, router modes, Configuration of Router: names, passwords, password encryption, interfaces name, creating a login banner, saving configuration and erasing configuration.

UNIT-II

Routing: Introduction, Types of routing: static and dynamic routing, Static routing: Configuring a static route, permanent keyword, static routes and administrative distance, configuring default route. Dynamic routing: Dynamic Routing Protocols: RIPv1, RIPv2, IGRP, EIGRP and OSPF. Delivery Semantics: unicast, anycast, multicast, broadcast, geocast. Configuring and Verifying RIP, IGRP, OSPF, EIGRP.

UNIT-III

Switch: introduction, roles of switches in network, types of switches: managed and unmanaged switch. Working of switch in different layers. Configuration of Switch: Command Mode, Setting Host Names, Setting Passwords, Setting IP Addresses and Default Gateways, Setting Operational Speed, Switch Port Security, Sticky MAC Addresses.

UNIT- IV

Switching: Introduction, Switching Services, Bridging vs LAN Switching, Switch Functions at Layer 2, Definition of Spanning Tree Protocol (STP), Operations of STP, STP Port States, Definition of VLAN, Typing of VLAN: Static and Dynamic, VLAN Identification Methods, VLAN Trunking Protocol (VTP), VTP Modes of Operation, Routing between VLANs, Configuring Inter-VLAN Routing.

Practical based on HWN-304

Practical: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
- Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
- Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
- Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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Paper Title: HWN-305-ROUTING CONFIGURATION & NETWORK CONNECTIVITY

Job Role: Network Support Engineer

Practical: 80
Internal Assessment: 20
Total Marks:100
Time:3 hours

Objective: Routing technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 10 marks.
- Practical file carries 10 marks and 40 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT- I

Functionality of Router and Switch: Describe basic routing concepts: packet forwarding, router lookup process, Process Switching/Fast Switching/CEF, Configure and verify utilizing the CLI to set basic Router configuration: hostname, local user and password, enable secret password, console and VTY logins, exec-timeout, service password encryption, Interface IP Address, loopback, banner, motd, copy run start.

UNIT-II

Configure and verify operation status of a device interface: Serial, Ethernet, Verify router configuration and network connectivity using, ping, extended, traceroute, telnet, SSH, sh cdp neighbors. Describe the boot process of Cisco IOS routers, Configure and verify utilizing the CLI to set basic Router configuration, Configure and verify operation status of a device interface, both serial and Ethernet, Verify router configuration and network connectivity. IP Routing Technologies: Configure and verify routing configuration for a static or default route given specific routing requirements.

UNIT-III

Differentiate methods of routing and routing protocols, Static versus Dynamic, 4.6.b Link state versus Distance Vector, next hop, ip routing table, Passive Interfaces (how they work), Admin Distance, and split horizon, metric. Configure and Verify OSPF: Benefit of single area, Configure OSPv2, Configure OSPv3, Router ID, Passive Interface, Discuss multi-area OSPF, Understand LSA types and purpose.

UNIT-IV

Configure and verify inter VLAN routing (Router on a stick), sub interfaces, upstream routing, encapsulation, Configure SVI interfaces, Manage Cisco IOS Files, Boot Preferences, Cisco IOS Images (15), Licensing, Show license, Change license, Configure and verify EIGRP (single AS), Feasible Distance/Feasible Successors/Administrative distance, Feasibility condition, Metric composition, Router ID, auto summary, Path Selection, Load Balancing, Unequal, Equal.

Books Recommended:

- Lacoste, CCNP Routing and Switching TSHOOT 300-135: Official Cert Guide, Pearson India
- Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
- Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

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SEMESTER-IV

Paper Title: GC 402 - PROJECT MANAGEMENT

Theory:80
Internal Assessment: 20
Total Marks : 100
Time:3 hours

Objectives: The objective of this paper is to provide knowledge to students about the essentials of undertaking projects in an organizational environment.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Concept of Project: Meaning, Characteristics, Classification of Projects, Project Life Cycle and its Phases. Project Manager: Roles and Responsibilities, Project Management as a Profession.

UNIT-II

Generating and Screening Ideas – Steps, Monitoring the Environment, Scouting for Project Ideas, Preliminarily Screening, Project Rating Index. Feasibility Studies – Technical, Financial, Economic, Social, Legal and Managerial.

UNIT-III

Project Appraisal Techniques: Objectives, Types and Methods. Project Risks: Meaning, Types, Measurement of Risk,. Decision Tree Analysis (Basic Concepts only).
Project Evaluation – Meaning, Evaluation v/s Appraisal, Objectives, Types of Evaluation, Techniques.

UNIT-IV

Project Organization and Control – Project Network Analysis (Basic concepts of PERT, CPM, Cost and Time Over Run). Project Reporting: Meaning, Purpose, Process, Requirements of a Good Report, Methods, Principles of Good Reporting System.

Books Recommended:

- Project Management - Choudhary - TataMcGraw Hill Pub.
- Project Management: The Managerial Process (Special Indian Edit.) -Clifford F Gray, Oregon State University.
- Projects : Planning, Analysis, Selection, Financing, implementation and Review - Chandra, Prasanna.

Paper Title: HWN-403- WINDOWS SERVER ADMINISTRATION

Job Role: Network Support Engineer

Theory:40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objective: The objective of this paper is to provide practical knowledge to the students about the Network Server.This course is meant to examine Windows Server while giving maximum learning potential to new Windows Administrators and Windows Developers.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Network Operating Systems: Windows Server Editions, Linux, Unix. Terminal services: Dumb terminals, terminal server configuration, windows server as terminal server, advantages of terminal services. Windows Server: different ways to install network operating system, pre-installation requirements. Maintaining AD DS, Managing User and Service Accounts, Implementing a Group Policy Infrastructure, Managing User Desktops with Group Policy, Installing, Configuring, and Troubleshooting the Network Policy Server Role.

UNIT-II

Installing and Configuring a Network Policy Server: Implementing Network Access Protection, Implementing Remote Access, Implementing Direct Access by Using the Getting Started Wizard Lab : Deploying an Advanced Direct Access Solution Lab : Implementing VPN Lab : Implementing Web Application Proxy, Optimizing File Services, Configuring Quotas and File Screening Using File Server Resource Manager Lab: Implementing Distributed File System, Configuring Encryption and Advanced Auditing, Deploying and Maintaining ServerImages, Using Windows Deployment Services to Deploy Windows Server, Implementing Update Management, Monitoring Windows Server.

UNIT-III

Microsoft Windows Server: Deploying and Managing Windows Server, Introduction to Active Directory Domain Services, Installing Domain Controllers, Managing Active Directory Domain Services Objects, Automating Active Directory Domain Services Administration, Implementing Dynamic Host Configuration Protocol, Implementing DNS, Implementing Local Storage, Implementing File and Print Services.

UNIT-IV

Windows Security: Firewalls, Spyware, viruses, User accounts security, Search strategies System performance, Troubleshooting tools, Windows registry. Administering Windows Server: Configuring and Troubleshooting Domain Name System, Configuring and Troubleshooting DNS, Maintaining Active Directory Domain Services.

Practical based on HWN-403

Practical: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Windows Server Administration Fundamentals by Microsoft Official Academic Course
- Darril Gibson, Microsoft Windows Networking Essentials 1st Edition
- Tom Carpenter, Microsoft Windows Server Administration Essentials
- Jamsa, Kris A., 1993: Rescued by Windows 3.1, Galgotia. 5. Basandra, S.K., 1995 Computers Today, Galgotia.

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Paper Title: HWN-404-NETWORK ADMINISTRATION & TROUBLESHOOTING

Job Role: Network Support Engineer

Theory:40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objective: The goal of this course is to train students to an exceptionally high standards in network planning, implementation, administration and support.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Backing UP and Restoring IOS Software and Configuration: Boot system commands, IOS file system, backing up configuration to a TFTP server, restoring configuration to a TFTP server, backing up the IOS software to a TFTP server, restoring/upgrading the IOS software from a TFTP server, restoring IOS software from ROM Monitor mode using Xmodem.

UNIT-II

Password-Recovery Procedures and configuration Register**:**The configuration register: A visual representation, bit means, the boot field, console terminal baud rate setting, console line speed: CLI & ROM Monitor mode, password recovery procedures for router, password recovery procedures for switches.

UNIT-III

Basic troubleshooting**:** viewing routing table, determining the gateway of last resort, last routing update, OSI layer 3 testing, OSI layer 7 testing, interpreting the show interface command, using CDP to troubleshooting, show controllers command debug commands, time stamps, Operating system IP verification commands, the ip http server command, netstat command.

UNIT-IV

Security Device Manager: SDM connecting with CLI and GUI, SDM Express Wizard with no CLI pre-configuration, resetting the router to factory defaults, SDM user interfaces, configuration routing using SDM, SDM monitor mode, SDM to configuration a router as a DHCP server & client, SDM configuration NAT/PAT.

Practical based on HWN-404

Practical: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Cisco Networking Essentials, Author: mcmillan, troy|author; John Wiley & Sons
- Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press
- Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media
- Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

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Paper Title: HWN-405-SWITCHING CONFIGURATION & NETWORK CONNECTIVITY

Job Role: Network Support Engineer

Practical:80
Internal Assessment: 20
Total Marks:100
Time:3 hours

Objective: Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 10 marks.
- Practical file carries 10 marks and 40 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT-I

IP Services: Configure and verify DHCP (IOS Router), Configuring router interfaces to use DHCP, DHCP options (basic overview and functionality), Excluded addresses, Lease time, Describe the types, features, and applications of ACLs, standard (editing and sequence numbers) extended, named, numbered, Log option. Configure and verify ACLs in a network environment, named, numbered, Log option, Identify the basic operation of NAT, purpose, pool, static, 1 to 1, overloading, source addressing, one way NAT, Configure and verify NAT for given network requirements.

UNIT-II

Configure and verify NTP as a client. Recognize High availability (FHRP), VRRP, HSRP, GLBP, and Configure and verify syslog, Utilize Syslog Output, Describe SNMP v2 and v3. LAN Switching Technologies: Determine the technology and media access control method for Ethernet networks, Identify basic switching concepts and the operation of Cisco switches: Collision Domains, Broadcast Domains, Ways to switch, Store, Forward, Cut through, CAM Table.

UNIT-III

Configure and verify initial switch configuration including remote access management: Hostname, management IP address, IP default-gateway, local user and password, enable secret password Console and VTY logins, exec-timeout service password encryption, copy run start, Verify network status and switch operation using basic utilities such as: ping, telnet, SSH.

Describe how VLANs create logically separate networks and the need for routing between them, Explain network segmentation and basic traffic management concepts, Configure and verify VLANs.

UNIT-IV

Configure and verify trunking on Cisco switches: dtp (topic), auto-negotiation, Identify enhanced switching technologies: RSTP, PVSTP, Ether channels, Configure and verify PVSTP operation: Describe root bridge election, Spanning tree mode. Configure and verify Switch Port Security: Sticky mac, MAC address limitation, static/dynamic, violation modes, err disable, shutdown, protect restrict, shutdown unused ports, err disable recovery, assign unused ports in unused VLANs, putting Native VLAN to other than VLAN 1.

Books Recommended:

- Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
- Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
- Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media

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SEMESTER-V

Paper Title: GC 502 - INTRODUCTION TO RESEARCH METHODOLOGY AND REPORT WRITING

Theory:80
Internal Assessment: 20
Total Marks : 100
Time:3 hours

Objective: The objective of the module is to familiarize the students with basics of research methodology like types of research, data collection, sampling techniques and report writing.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Research - Meaning, Characteristics, Types, Process and Utility,. Research Design - Meaning, Types and Features of good research design. Qualitative and Quantitative research. Importance of review of literature. The research proposal.

UNIT II

Primary and Secondary data. Methods of data collection. Measurement and scaling, Designing of Questionnaire and schedule.

UNIT III

Sampling Design and Sampling procedure. Sampling and non-sampling errors. Data analysis and interpretation. Role of computers in research.

UNIT IV

Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals; Case Studies.

Books Recommended:

- 'Research Methods and Techniques' by C.R. Kothari, Eastern Wiley Publications.
- 'Business Research Methods' by A. Bryman, and E. Bell, Oxford University Press, New York.
- 'Research Methodology A Step by Step Guide for Beginners' by R. Kumar, Pearson Education.

Paper Title: HWN-503- LINUX OPERATING SYSTEM

Job Role: System Administrator

Theory: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objectives: This course enables students to get familiar with Linux system, its commands, files & directories, shell programming, and system administration. Student will learn to Work in the Linux environmentand administrate the Linux machine.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT – I

Introduction: Linux Operating System, features, Basic Architecture, features of Kernel and Shell. Linux File system - Boot block, super block, Inode table, data blocks, How Linux kernel access files, Linux standard file system. Structure of file system, Essential Linux commands - Commands for files and directories : cat, cd, ls, cp, md, rm, mkdir, rmdir, pwd, file, more, less, file comparisons – cmp & comm, View files, disk related commands, checking disk free spaces, chmod with its options, cal,date,who,tty, lp,stty.

UNIT –II

Filters and pipes: head, tail, wc, pr, cut, paste, sort, uniqe, grep, egrep, fgrep, tee. The process : shell process, parent and children ,process status, system process, multiple jobs in background and foreground, changing process priority with nice, premature termination of process, Mathematical commands- bc, expr, factor, units.

UNIT –III

Creating and editing files with VI editor with their command options, Operators, text deletion, text movement, changing text, yanking text, filtering text, the ex-mode, moving text from one file to another. Communication: The bulletin board –news, write, mesg, talk, mail, elm, pine, finger, vacation and connecting to remote machine.

UNIT -IV

System administration Common administrative tasks, identifying administrative files – configuration and log files, Role of system administrator, Managing user accounts-adding &deleting users, changing permissions and ownerships, Installation of Linux system– Linux Installation requirement, complete Procedure steps, Partitioning the Hard drive, System startup and shut-down process, init and run levels. File system mounting, lpstat, backup strategy, installing software on Linux.

Practical based on HWN-503

Practical: 40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.

- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Tim Parker: Linux Unleashed, Techmedia Publishing House.
- Norton, P.: Complete guide to LINUX, Techmedia.
- Komarinski, M.: LINUX System Administration Handbook, AW.
- John Goerzen: Linux Programming Bible, IDG Books, New Delhi.
- Venkateshmurthy, M.G.: Introduction to UNIX & Shell Programming, Pearson Education.
- Tery Dawson, Gregor N. Purdy, Linux Network Administration Guide, Oreilly.

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Paper Title: HWN-504- FIREWALL TECHNOLOGY

Job Role: System Administrator

Theory: 40
Internal Assessment: 10
Total Marks:50
Time:3 hours

Objectives: This paper enable the students to learn to configure, install and administrate a firewall and handle functionality of firewall.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT-I

Firewall: Introduction, Why do we need Firewall, How Firewall works, Network Layer Firewall, Application Layer Firewall, Windows Firewall, Types of firewall: Hardware and Software Firewalls, Firewall Filtering Techniques: Packet Filter, Application Gateway, Circuit-level Gateway, Proxy Server. Installation of Firewall: key functions during installation, special options during installation, installing UTM, Basic Configuration, Backup Restoration.

UNIT-II

Web Admin: Web Admin Menu Button bar, lists searching in lists, dialog boxes, buttons and icons, object lists. Dashboard: dashboard settings, flow monitor, Management: System settings, organizational, hostname, time and date, scan settings, reset configuration or passwords. Web admin settings: general, access control, user rights HTTPs certificate, user preferences, Licensing, up2date, backup/restore, user portal, notifications, hardware and software requirements, status system status configuration, shutdown and restart.

UNIT-III

Definitions & Users: Network definitions, MAC address definitions, service definitions, time period definitions, users and groups, global settings Interfaces and routing, Network Services: DNS: global, forwarders, request routing, static entries, DynDNS, DHCP: Servers, Relay, DHCPv6 Relay, Static Mappings, IPv4 table, Network Protection: Firewall, Rules, Country Blocking, Country Blocking Exceptions, ICMP, Advanced, NAT: Masquerading, NAT, Advanced Threat Protection: Global, Intrusion Prevention: Global attack patterns, Anti-Dos/Flooding, Anti-Portscan, Exceptions, Advanced, Server Load Balancing: Balancing Rules

UNIT-IV

Web Protection: Web filtering: some key differences, common tasks, migration, Global, Https, Policies: Filter action wizard, categories, websites, downloads, antivirus, Web filter Profile: Filter profiles, filter actions, parent proxies, Filtering options: exceptions, websites, bypass users, potentially unwanted applications categories, Logging & Reporting : View Log Files, Today's Log Files, Archived Log Files, Search Log Files, Hardware Daily, Weekly, Monthly, Yearly.

Practical based on HWN-504

Practical: 40
Internal Assessment: 10
Total Marks: 50
Time:3 hours

Instructions:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 5 marks.
- Practical file carries 5 marks and 20 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

Books Recommended:

- Hucaby, "CCNP Security FIREWALL 642-618 Official Cert Guide", Pearson India
- **Burns, "CCNP Security IPS 642-627 Official Cert Guide", Pearson India**
- **Wilkins, "CCNP Security Secure 642-637 Official Cert Guide", Publisher: Pearson India**
- **Dameon D Welch-Abernathy, "Essential Check Point Firewall-1 NG: An Installation, Configuration, and Troubleshooting Guide", Pearson Education**
- **"Key Features of Framework for Monitoring Hardware Firewall Functionality Using Nids", LAP Lambert Academic Publishing.**

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Paper Title: HWN-505- ADVANCED ROUTING AND SWICHING

Job Role: System Administrator

Practical: 80
Internal Assessment: 20
Total Marks:100
Time:3 hours

Objective: Routing and Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.

Instruction:

- Examiner will set total of four questions covering the whole syllabus.
- Student will attempt any three question
- Each question carries 10 marks.
- Practical file carries 10 marks and 40 marks for practical work and viva-voce.
- Practical paper will be conducted by the college in collaboration with the respective experts and marks will be submitted to Panjab University.

UNIT- I

IP Addressing (IPv4/IPv6): Describe the operation and necessity of using private and public IP addresses for IPv4 addressing, Identify the appropriate IPv6 addressing scheme to satisfy addressing requirements in a LAN/WAN environment.

UNIT- II

Identify the appropriate IPv4 addressing scheme using VLSM and summarization to satisfy addressing requirements in a LAN/WAN environment. Describe the technological requirements for running IPv6 in conjunction with IPv4, dual stack, Describe IPv6 addresses, global unicast, Multicast, link local, unique local, eui 64, auto-configuration.

UNIT- III

Network Device Security: Configure and verify network device security features: Device password security, enable secret versus enable, Transport, disable telnet, SSH, VTYs, physical security, service password.

UNIT- IV

Describe external authentication methods. Configure and verify a basic WAN serial connection, Configure and verify a PPP connection between Cisco routers, Configure and verify Frame Relay on Cisco routers, Implement and troubleshoot PPPoE.

Books Recommended:

- Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India
- Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd
- CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education
- Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons
- Crypto and Network Security, 3rd Edition, Mcgraw Hill Education
- Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media.

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SEMESTER-VI

Paper Title: GC 602 - TOTAL QUALITY MANAGEMENT

Theory:80
Internal Assessment: 20
Total Marks : 100
Time:3 hours

Objective: The objective of the paper is to provide knowledge to students of the concepts of total quality management and to inculcate among them a concern for quality and customer satisfaction.

Instructions:

- The syllabus of this paper has been divided into FOUR units.

- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

UNIT I

Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis, Basic concepts of Total Quality Management, Principles of TQM.

UNIT II

Quality Council and Statements, Barriers to TQM Implementation. Quality: Customer Perception and Satisfaction, Customer Complaints, Service Quality, Customer Retention, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT III

Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process, Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM)

UNIT IV

Latest Challenges of Quality, Six Sigma Concepts: application. Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System– Elements, Implementation of Quality System, Documentation, Quality Auditing; ISO 14000, Requirements and Benefits.

Books Recommended:

- 'Total Quality Management' by A.V. Feigenbaum, McGraw-Hill Publication.
- 'Total Quality Management' by J.S. Oakland, Butterworth – Hcinemann Ltd., Oxford Press.

- 'Quality Management – Concepts and Tasks' by V. Narayana & N.S.Sreenivasan, New Age International.
- Total Quality Management for Engineers' by Zeiri, Wood Head Publishers.

'Total Quality Management' by James R. Evans, Cengage Learning

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Paper Title: HWN- 603-INDUSTRIAL TRAINING

Job Role: System Administrator

Total Marls: 300

Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce. These credits will be evaluated in Semester VI

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